Ex: 12(b) | Implement a chat program using Socket

Aim:

To simulate End-end communication at transport layer to implent a chat program using client.

```
import socket
import threading
def udp_server():
    s = socket.socket(socket.AF_INET)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == "quit":
            print(f"[SERVER] client {addr}
                ended the connection")
            break
        print(f"[SERVER] Recived from {addr}
            : {msg}")
        s.sendto(data, addr)
    s.close()
    print("[SERVER] Stopped")
```

Client:

```
def udp_client():
    c = socket.socket(socket.AF_INET)
    server = ('127.0.0.1', 12000)
while True:
    msg = input("Enter message (or 'quit' to stop): ")
    c.sendto(msg.encode(), server)
    if msg.lower() == 'quit':
        print("[CLIENT] Closing connection")
    data, _ = c.recvfrom(1024)
    print("Echo from Server", data.decode())
c.close()
print("[CLIENT] Stopped.")
```

Main:

```
if __name__ == "__main__":
    threading.Thread(target = udp_server, True).start()
    udp_client().
```

Output:

```
Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1:12000
hi
[SERVER] Received from ('127.0.0.1:12000') hi
Echo from Server: hi
Enter message (or 'quit' to stop): hello
[SERVER] Received from ('127.0.0.1:49670): hello
Echo from Server: hello
Enter message (or 'quit to stop): quit
[CLIENT] Closing connection....
[SERVER] end the connection
[CLIENT] STOPPED [SERVER] Stopped.
```

Result:

This implementation of echo client server using TCP/UDP is done successfully.